

AstroLight: A Late-Fusion Deep Learning Architecture for Multi-Class Exoplanet Transit Signal Classification

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Abstract—Automated classification of Kepler Objects of Interest (KOIs) into physically motivated false-positive categories is a critical bottleneck in the exoplanet vetting pipeline. While prior deep learning systems (AstroNet, ExoMiner) excel at binary planet-candidate vs. false-positive discrimination, the multi-class problem has received limited attention. We present AstroLight, a multi-task late-fusion architecture that classifies KOIs into five categories: genuine transit, stellar eclipse, centroid offset, background noise, and ephemeris match. Developed through two model generations, our system evolves from a 1D-CNN BiLSTM hackathon prototype (44.3% accuracy, AUC 0.589) to a three-branch residual late-fusion model trained with Focal Loss that achieves 79.9% five-class accuracy and a weighted F1 of 0.804. We provide a systematic investigation of loss-function design under extreme class imbalance and a rigorous failure analysis of the centroid-offset over-prediction bias and the ephemeris-match data starvation problem.

Index Terms—Exoplanet detection, Kepler mission, multi-class classification, convolutional neural networks, focal loss, light-curve analysis, false-positive vetting.

I. INTRODUCTION

The NASA Kepler mission (2009–2018) observed 150,000+ stars, generating a rich archive of light curves from which ~9,000 planet candidates were extracted via the transit method. Distinguishing genuine planetary transits from astrophysical false positives — eclipsing binaries (EBs), background contamination, centroid offsets and instrument artefacts — remains a fundamental challenge requiring labour-intensive expert review.

Traditional automated vetting used the rule-based Robovetter pipeline [3], which lacked robustness across stellar populations. Machine learning approaches, beginning with the AstroNet CNN [1] (97.5% binary accuracy), demonstrated that phase-folded light curves contain sufficient morphological information for automated discrimination. However, all major published systems frame vetting as a *binary* problem, conflating physically distinct false-positive mechanisms into a single label.

This paper presents **AstroLight**, developed initially as an ISRO hackathon prototype and subsequently extended into a full research system. Our primary contributions are: (1) a five-class vetting taxonomy applied to the full Kepler KOI cumulative catalog; (2) a three-branch late-fusion architecture jointly processing global/local light curves and stellar catalog features; (3) a multi-task loss combining classification, parameter regression, and confidence estimation; and (4) empirical analysis of training instabilities caused by stacking multiple imbalance-correction mechanisms.

II. RELATED WORK

A. Classical Vetting

Robovetter [3] applied fixed threshold rules to TCE diagnostic metrics across Kepler data releases. While interpretable, its per-release manual calibration did not generalise across stellar population diversity.

B. Binary Deep Learning Classifiers

Shallue & Vanderburg [1] established the dual-view paradigm: a global phase-folded view capturing overall morphology and a local view centred on the transit dip, each processed by a 1D-CNN. AstroNet achieved 97.5% precision on Kepler DR24, setting a strong binary baseline. Ansdell et al. [4] extended this to DR25 with transfer learning. Yu et al. [5] incorporated stellar parameters, anticipating multi-modal fusion. ExoMiner [2] combined multiple diagnostic time series in a hierarchical CNN, reaching AUC 0.984 on binary classification.

C. Multi-Class Approaches

Dattilo et al. [6] explored three-class discrimination (PC, EB, BEB) on TESS data. Five-class classification with ephemeris matching as a distinct category has, to our knowledge, not been systematically reported. Direct comparison with [1, 2] is therefore qualitative; our task is strictly harder.

TABLE I

Comparison with Published Methods (* = binary classification)

Method	Cls	Acc.	AUC/W-F1	Inputs
AstroNet [1]	2*	97.5%	0.960	G+L
ExoMiner [2]	2*	~98%	0.984	Multi
Ansdell [4]	2*	97.8%	0.950	G+L
Dattilo [6]	3*	88.4%	0.843	G+L
Ours – M1	5	44.3%	0.589	G
Ours – M2 CE	5	72.1%	0.752	G+L+S
Ours – M2 FL	5	79.9%	0.804	G+L+S

III. DATASET AND PREPROCESSING

A. NASA Kepler KOI Catalog

Our dataset derives from the NASA Kepler Objects of Interest cumulative catalog with manually-curated five-class signal labels. Each of 9,564 entries links a Kepler Input Catalog (KIC) identifier to a downloaded .npz light curve file. The class distribution is severely skewed (Table II); Ephemeris Match constitutes only 1.3% of samples.

TABLE II
Dataset Class Distribution

Class	Train	Val	Test	%
Transit	3307	708	709	49.4
Stellar Eclipse	1547	331	331	23.1
Not Transit	909	196	195	13.6
Centroid Offset	844	182	181	12.6
Ephemeris Match	87	22	19	1.3
Total	6694	1439	1435	100

B. Preprocessing Pipeline

Raw light curves undergo: (1) 5-sigma outlier clipping; (2) phase-folding to $\phi \in [-0.5, 0.5]$; (3) global view: 1,001-bin median representation over full phase range; (4) local view: 1,001-bin representation over $[-W, W]$, $W = 2.5 \times (T_{\text{dur}}/P)$; (5)

per-view median-subtraction and std-normalisation. Twelve stellar catalog features (koi_period, koi_depth, etc.) are Z-score normalised using training-split statistics; NaN values are imputed with per-column training medians. Preprocessing results are cached after epoch 1, reducing per-epoch loading from 166 s to ~14 s.

Training samples receive stochastic augmentation applied after cache retrieval: additive Gaussian noise ($\sigma=0.008$, $p=0.5$), circular phase shift (± 30 bins, $p=0.5$), and multiplicative scaling ($[0.98, 1.02]$, $p=0.5$).

IV. MODEL ARCHITECTURES

A. Model 1: CNN–Attention–BiLSTM (Hackathon Prototype)

Model 1 (ExoplanetDeepModel) was developed under hackathon constraints for the ISRO Space Tech challenge as a proof-of-concept five-class classifier operating on the *global* light curve only. The architecture comprises four sequential stages:

- **Stage 1 – 1D CNN Extractor:** Three ConvBlock1D layers with residual connections, channels [32, 64, 128], kernels [15, 11, 7], max-pool-4 at final layer. Output: (B, 128, 500) for input length 2,000.
- **Stage 2 – Temporal Attention:** Multi-head self-attention (4 heads, dim=128) applied along the temporal axis. Attention weights provide interpretable saliency maps.
- **Stage 3 – Bidirectional LSTM:** 2-layer BiLSTM (hidden=128), output globally average-pooled to (B, 256).
- **Stage 4 – Shared Embedding:** Linear 256→128 + BatchNorm + ReLU + Dropout(0.3). Three multi-task heads (classification 128→5, regression 128→4, confidence 128→1).

Model 1 was trained with AdamW ($\text{lr}=2\text{e-}4$), FocalLoss ($\gamma=2.0$) with inverse-frequency class weights, and a per-parameter regression loss ($L_{\text{total}} = L_{\text{cls}} + 0.1 \blacksquare L_{\text{depth}} + 0.1 \blacksquare L_{\text{dur}} + 0.1 \blacksquare L_{\text{per}} + 0.1 \blacksquare L_{\text{mid}} + L_{\text{conf}}$). Masked regression

(transit samples only) prevented gradient corruption from non-transit geometry. Due to hackathon time constraints, only the global view was used; the local view and stellar feature branch were not implemented.

B. Model 2: Late-Fusion Multi-Task Architecture

Model 2 (ExoplanetLateFusionModel) addresses Model 1's limitations through three parallel feature extraction branches:

- **Global Branch (GlobalViewBranch):** 1D-ResNet applied to the 1,001-bin global view, producing (B, 32, 256). Global Average Pooling → 256-d embedding.
- **Local Branch (LocalViewBranch):** Identical architecture applied to the local transit view. Output: 256-d embedding.
- **Stellar Branch (StellarFeaturesBranch):** 3-layer FC network (12→64→128), BatchNorm. Output: 128-d embedding.

The three embeddings are concatenated into a 640-d joint representation and refined by a **ResidualLateFusionHead**: MLP (640→512→256) with a parallel shortcut (640→256), BatchNorm, ReLU, Dropout(0.4). The 256-d output feeds three heads identical in structure to Model 1. A learnable temperature parameter scales classification logits before softmax.

V. TRAINING STRATEGY

A. Model 2 Optimisation

AdamW (lr=1e-3, λ=1e-4) with OneCycleLR cosine annealing (max_lr=3e-3, pct_start=0.15, 80 epochs). Gradient norm clipped to 5.0. Early stopping on val ROC-AUC (patience=20). A WeightedRandomSampler draws training batches using inverse-frequency weights with ephemeris_boost=2.0× for class 4.

B. Loss Function Comparison

Run 1 – CrossEntropy (CE): Standard weighted CE with full inverse-frequency class weights. Converged at epoch 34, val AUC=0.8992.

Run 2 – Focal Loss (FL): Replaces CE with $FL(p_i) = -(1-p_i)^\gamma \log(p_i)$, $\gamma=2.0$. Class weights are removed from the loss because WeightedRandomSampler already balances the data distribution; passing class weights into FL additionally caused model collapse (§VI-B).

VI. EXPERIMENTS AND RESULTS

A. Model 1 Results

Model 1 achieved 44.3% overall accuracy and AUC 0.589 on the five-class test set. Transit recall was high (85.7%) — the model strongly anchored to the majority class — but precision was poor (42.5%). Centroid Offset and Not Transit were nearly undetected, and Ephemeris Match F1 was 0.00. These results confirm that processing only the global light curve without stellar features is insufficient for multi-class discrimination.

B. Model 2 Results

TABLE III

Per-class performance on test set (1,435 samples). M1 = Model 1 (CNN-BiLSTM); CE = Model 2 + CrossEntropy; FL = Model 2 + Focal Loss (best).

Class	M1 F1	CE Prec	CE Rec	CE F1	FL Prec	FL Rec	FL F1	N
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Transit	0.568	0.970	0.784	0.867	0.915	0.937	0.925	1,709
Stellar Ecl.	0.484	0.854	0.653	0.740	0.843	0.665	0.743	1,881
Not Transit	0.098	0.662	0.703	0.682	0.718	0.692	0.705	1,195
Centroid Off.	0.031	0.364	0.685	0.475	0.565	0.690	0.624	1,181
Ephem. Match	0.000	0.033	0.105	0.050	0.054	0.105	0.071	19
Accuracy	0.443			0.721			0.799	1,435
Macro F1	0.368			0.563			0.614	1,435
Weighted F1	0.368			0.752			0.804	1,435
Val AUC	0.589			0.899			~0.90+	—

VII. FAILURE ANALYSIS

A. Centroid Offset Gravity Well

In Run 1 (CE), Centroid Offset precision was 0.364 — 64% of centroid_offset predictions were incorrect. Inspection revealed the model was using this class as a *default rejection bucket*: ambiguous signals with no strong class evidence defaulted to centroid_offset. This directly suppressed Transit recall to 78.4%, as one in five real transits was misclassified as centroid_offset.

Focal Loss resolves this by scaling gradients as $(1-p_i)^2$: overconfident wrong predictions receive near-zero gradient, while hard misclassifications dominate training. After switching to FL, Centroid Offset precision improved from 0.364 to 0.565 (+55% relative), and Transit recall recovered to 93.7%.

B. Loss-Stacking Collapse

When Focal Loss was introduced while retaining full inverse-frequency class weights, a catastrophic collapse occurred: the model defaulted to predicting Ephemeris Match for ~95% of inputs (overall accuracy 24.4%). The combined effect of WeightedRandomSampler (×38 for Ephemeris vs Transit) × inverse-frequency weights × Focal Loss created an

effective $\sim 120\times$ gradient amplification for the rarest class, rendering all other classes' loss signals negligible. Resolution: remove class weights from FL entirely, relying on the Sampler for data-level balance and FL for hard-example focus — orthogonal mechanisms that do not compound.

C. Epheris Match: Data Starvation

Epheris Match achieves $F1 \leq 0.07$ across all configurations. With only 87 training samples, the model cannot learn reliable decision boundaries. Critically, Epheris Match is *morphologically indistinguishable* from genuine transit in the light curve: the discriminating signal is the period's coincidence with a catalog entry, encoded weakly in `koi_period` — not a direct photometric signature.

An experiment with `ephemeris_boost=8.0 $\times$` and a selective loss weight of $5.0\times$ for class 4 improved Epheris F1 from 0.071 to 0.113 but degraded Stellar Eclipse recall from 66% to 55% — a net loss given Stellar Eclipse has $17\times$ more test samples. Conclusion: loss engineering cannot substitute for labeled data when class frequency drops below $\sim 1\%$.

VIII. DISCUSSION

- **One correction per axis:** Sampler (data-level) and weighted loss (gradient-level) correct the same underlying imbalance. Stacking both multiplicatively destabilises training.
- **Focal Loss > Weighted CE for multi-class overlap:** When classes share morphological features, FL's confidence-based modulation is more principled than fixed inverse-frequency weights.
- **Multi-task heads as regularisers:** Masked regression prevents classification from collapsing by enforcing physical consistency on transit geometry.
- **Local view contribution:** Model 2 (using both global + local views) improves Transit F1 from 0.568 (M1, global only) to 0.925, confirming that fine transit structure is critical for discriminating real transits from EBs.
- **Stellar features resolve centroid ambiguity:** The inclusion of `koi_fwm_stat_sig` (centroid shift significance) and `koi_dicco_msky` in the stellar branch provides direct physical evidence for centroid offset, reducing its over-prediction.

IX. CONCLUSION

We presented AstroLight, a two-generation deep learning system for five-class exoplanet transit signal discrimination on the NASA Kepler KOI catalog. Model 1 (CNN–Attention–BiLSTM, 44.3% accuracy) established a baseline showing that global light curves alone are insufficient. Model 2 (three-branch late-fusion + Focal Loss, 79.9% accuracy, W-F1=0.804) substantially improved all four major classes, with Transit F1 reaching 0.925 and Centroid Offset F1 improving from 0.031 to 0.624. The Epheris Match class remains near-undetectable ($F1=0.07$), representing a fundamental data limitation rather than a modelling failure. Our systematic investigation of loss-stacking collapse and the gravity-well effect provides practical guidance for multi-class imbalanced learning in astronomical classification tasks.

Future Work

- Catalog cross-matching feature: a hard binary indicator for period coincidence with known KOIs would directly encode epheris match evidence.
- Transformer-based encoder for long-range transit morphology modelling.
- TESS generalisation study across mission boundaries.
- Full ablation study quantifying each branch's marginal contribution.

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